

TELANGANA STATE - 2017
CLASS 10
GENERAL SCIENCE, Paper-II
(Biological Science)

Time: 2 Hrs. 45 min.

[Maximum Marks: 40]

Instructions:

1. **Part-‘A’** consist of three Sections. Answer the questions under **Part-‘A’** on a separate answer book.
2. Write the answers to the questions under **Part ‘B’** on the question paper itself and attach to the answer book of **Part ‘A’**.
3. 15 minutes is allotted to read the question paper. 2.30 hours is allocated to write answers.

Part-A

Time: 2 hrs.

Marks: 30

SECTION - I

Note:

- (i) Answer the following questions. 4x1=4
- (ii) Each question carries ONE mark.

1. Which disease occurs is called when there is an immediate second pregnancy or repeated child births in a mother?

Ans: Preeclampsia

2. What happens if blood platelets are absent in blood?

Ans: The blood platelets are essential for the process of blood clotting. Whenever an injury occurs the chances of blood loss

increases. In order to prevent the loss of blood it is necessary to form a clot.

3. What questions you ask the doctor, who visited your school on World AIDS day?

Ans: Firstly, the basic questions, what is AIDS and how did it transmit?

4. Suggest any two practices suitable to farmer with less water resources.

Ans: Drip irrigation and sprinkler system and are the two practice which is suitable to farmer with less water resources.

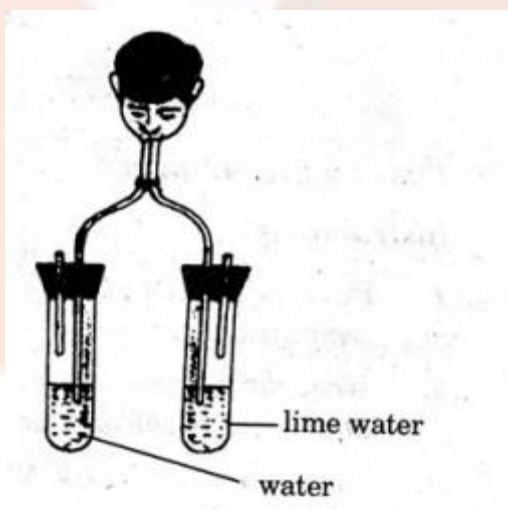
SECTION – II

NOTE:

- (i) Answer the following questions.
- (ii) Each question carries **TWO** marks.

5x2=10

5.



(a) Which gas turns lime water milky in this experiment?

Ans: Co₂

(b) Which gas do you think might be present in less quantities in the air we breathe out as compared to air around us?

Ans: Nitrogen

6. After reading the functions of lymphatic system, what precautions you would suggest to your elders about Edema?

Ans: After reading the functions of lymphatic system, Some of the precautions that I would suggest are listed below -

1. Reducing the intake of sodium especially common salt.

2. Prevention from staying in extreme temperatures.

3. Avoid wearing tight clothes so that it does not harm the affected area (if anyone suffers from this disease).

7. What habits you would like follow for the proper functioning of kidneys?

Ans: For the proper functioning of kidney I would like to follow the following:

- Staying hydrated
- Eating healthy foods
- Exercising regularly
- No smoking

8. It is believed that the Diencephalon in fore-brain and vagus nerve (10th cranial nerve) plays an important role in carrying hunger signals to the brain. Hunger pangs continue upto 30-45

minutes. Increase in ghrelin levels results in sensation of hunger and motivation to consume food.

Read about content and prepare any two questions.

Ans: Ques. 1. Hunger signals are produced by the which hormone in human body?

Ques. 2 What causes hunger pangs?

9. Explain the role of Green house gases in Global Warming.

Ans: Greenhouse gases act like a blanket, absorbing IR radiation and preventing it from escaping into outer space. The net effect is the gradual heating of Earth's atmosphere and surface, a process known as global warming.

SECTION - III

NOTE:

- (i) Answer the following questions. 4x4=16
- (ii) Each question carries **FOUR** marks.
- (iii) There is internal choice for each question.

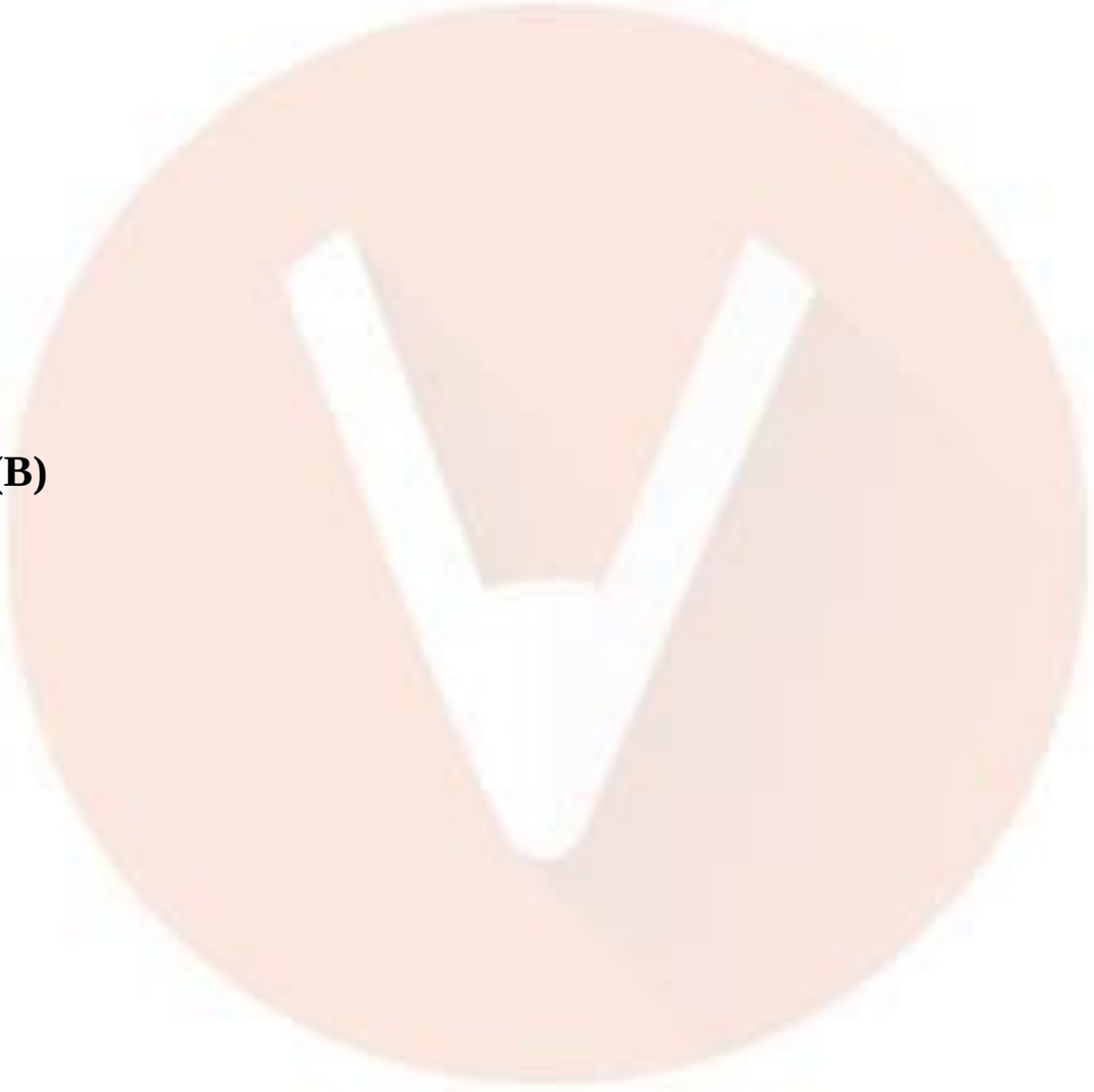
10. (A) During Photosynthesis, several events occur in the Chloroplast. Explain the light dependent reactions.

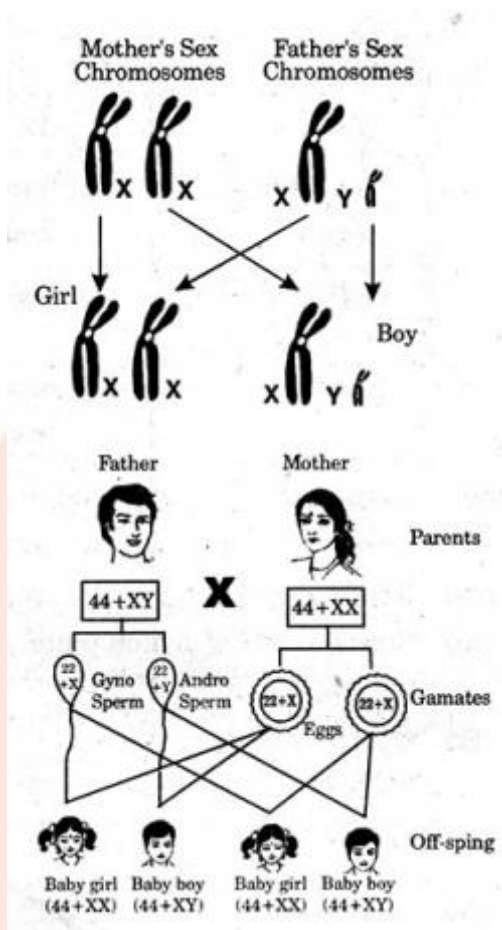
Ans: The chloroplast is known to pass through two different phases of photosynthesis; thylakoid is the place where the light reactions occur when water (H_2O) is oxidized and oxygen (O_2) is released. Then comes the turn of the light-independent or the "dark" reactions, which are performed in the chloroplast stroma. This ingenious process, aka carbon fixation, generates energy from ATP and NADPH molecules, this is possible when light

reactions give chance to a chemical (which utilizes carbon dioxide from the surrounding) to formulate glyceraldehyde-3-phosphate (G3P), a three-carbon sugar molecule.

OR

(B)





(i) What does the given flow chart indicate?

Ans. Sex determination system.

(ii) What will happen if the sperm containing 'X' chromosomes fertilize the ovum?

Ans. Give birth to female child.

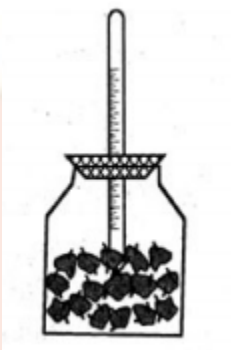
(iii) Who decides the sex of the baby-Mother or Father?

Ans. Father (XY)

(iv) How many pairs of chromosomes are present in off-spring?

Ans. 23 pairs.

11.



(A)

- (i) What change did you observe in the thermometer in the given experiment?

Ans. Rise in temperature (heat generation).

- (ii) Where does the heat come from?

Ans. From Germination process.

- (iii) What result you will get, if you perform this experiment with dry seeds?

Ans. There will be no change in the temperature

- (iv) What are the apparatus used in this experiment?

Ans. Thermometer, thermos flask and germinating seeds.

OR

(B)

- (i) Explain the procedure followed in the experiment conducted to understand the action of saliva on starch.

Ans:

Taking a test tube half-filled with water and adding a pinch of flour to it, shaking it well till it gets completely mixed.

Testing the presence of starch: Putting few drops of this mixture in a watch glass and putting a drop of dilute tincture iodine in it. If the color turns blue-black it confirms the presence of starch.

Now, dissolving a pinch of flour into half-filled water in a test tube and dividing the mixture into two equal halves.

Now, adding a teaspoon of saliva to one of the test tubes and marking it.

Now, after about 45 minutes, adding a drop of dilute tincture iodine solution to test tubes containing the solution.

The solution of test tube will show color change as starch is converted to sugar and there will be no color change in other test tube.

- (ii) What apparatus and chemicals are used to do this experiment?

Ans: Three test tubes, water, flour, dilute tincture, saliva is used.

12. (A) Analyze the following information and answer the questions.

Alkaloid	Part of the plant	Uses
Quinine	Bark	Anti-malarial drug
Pyrethroids	Leaves	Insecticide
Reserpine	Roots	Medicine for snake bite
Caffeine	Seeds	Central nervous system stimulant
Nimbin	Seeds, Barks, Leaves	Antiseptic

(i) Name the alkaloid which is used to cure malaria.

Ans: Quinine

(ii) Name the alkaloids used as insecticides.

Ans: Pyrethroids

(iii) Which system is stimulated by the alkaloid caffeine?

Ans: Central nervous system

(iv) Which parts of which plant is used as medicine for snake bite?

Ans: Roots of Reserpine

OR

(B) Should we use pesticides as they prevent our crop and food from pests or should we think of alternatives? Write your view about the issue and give sound reason for your answer.

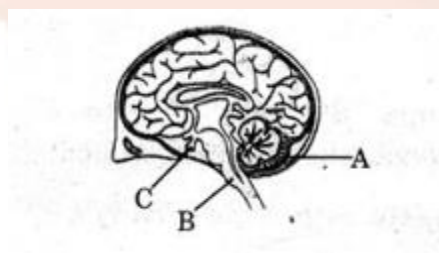
Ans: We should not use pesticides as they are chemicals and chemicals in pesticides could cause harm to human beings, animals and desired insect species.

We could use alternatives:

Biocontrol: It can be used in the form of birds, animals, insects which can eat up the insects and use of pesticides are not required.

2. Polyculture: Growth of only one type of plant could be dangerous for the colonization of the insects. So, Polyculture is found helpful in preventing the disease in plants.

13. (A)



(i) This diagram belongs to which system of the body?

Ans. Neurological system.

(ii) Name the parts A and B.

Ans. Part A is Cerebrum, and part B Spinal cord

(iii) The part 'C' is endocrinal gland. This is called master gland. What is the name of this gland?

Ans. Pituitary gland

(iv) Which part in this diagram is useful to solve problems and puzzles?

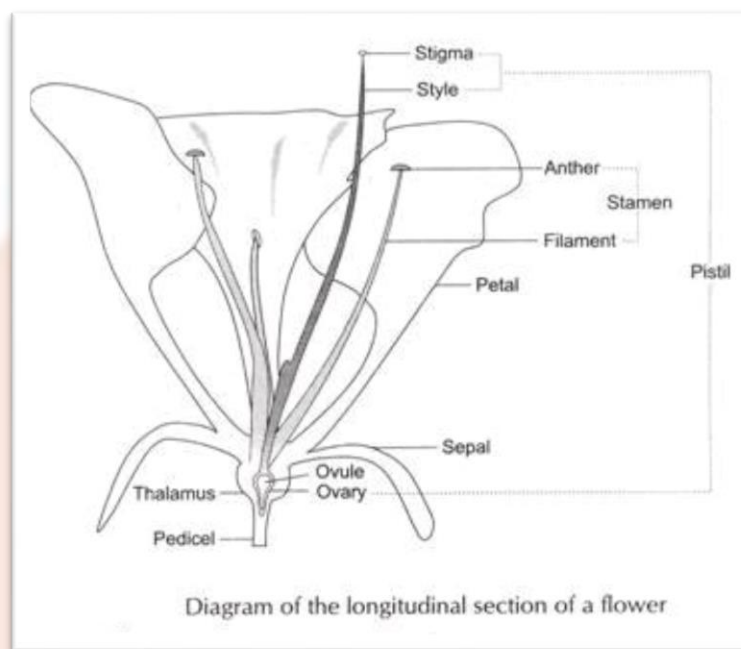
Ans. Cerebrum

OR

(B)

(i) Draw a neat labelled diagram of L.S. of flower?

Ans.



(ii) What are the sexual parts in the flower?

Ans. Stamen and pistil

Part-B**Time: 30 minutes****Marks: 10****SECTION - IV****Note:**

- (iii) Answer all the questions.
- (iv) Each question carries $\frac{1}{2}$ marks.
- (v) Answer should be written on question paper itself.

Put correct answer in the brackets given below. $20 \times \frac{1}{2} = 10$

14. Pepsin : Proteins :: Lipase : []

- (A) Carbohydrates
- (B) Fats
- (C) Vitamins
- (D) Sucrose

Ans – (B)

15. $C_6H_{12}O_6 + 6O_2 \rightarrow \dots\dots\dots + 6H_2O$ energy []

- (A) $6CO_2$
- (B) $C_6H_{12}O_6$
- (C) $6O_2$
- (D) $12CO_2$

Ans – (A)

16. 1. Chlorophyll is similar to the haemoglobin of blood. []

2. Haemoglobin has iron while chlorophyll has magnesium.

- (A) 1 is correct, 2 is wrong.
- (B) 1 is wrong, 2 is correct.
- (C) 1, 2 both are correct.
- (D) 1, 2 both are wrong.

Ans – (A)

17. The percentage of Carbon dioxide in exhaled air is
..... []

- (A) 44
- (B) 4.4
- (C) 0.4
- (D) 0.04

Ans – (D)

18. The given blood vessel carries blood from body parts to heart. Name the blood vessel. []



- (A) Artery
- (B) Capillary
- (C) Vein
- (D) Muscular fibre

Ans – (C)

19. Study the following pairs.

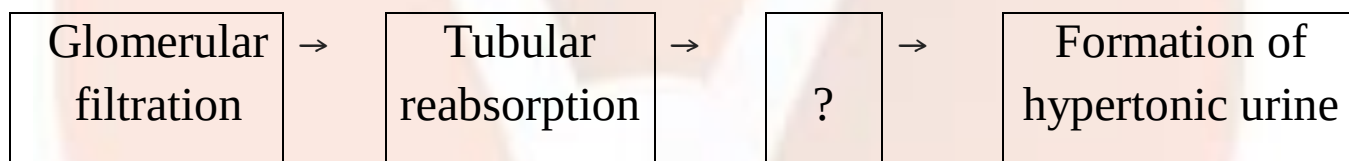
- (A) Stethoscope - Rene Laennec
- (B) Blood pressure - Thermometer
- (C) Amoeba - Brownian movement

Identify the mis-matched pair.

- (A) a
- (B) b
- (C) c
- (D) None

Ans – (B)

20. Fill the flow chart of urine formation.



- (A) Tubular secretion.
- (B) Tubular filtration.
- (C) Tubular excretion.
- (D) Urine formation.

Ans – (A)

21. I am a plant. My seeds are used for production of bio-fuel.

Who am I?

[]

- (A) Rubber plant
- (B) Neem plant

- (C) Cactus plant
- (D) Jatropha plant

Ans – (D)

22. The excretory organs in phylum Mollusca is []

- (A) Renette cells.
- (B) Green glands.
- (C) Meta nephridia.
- (D) Kidneys.

Ans – (C)

23. Leaf movement in mimosa helps to []

- (A) reduce photosynthesis.
- (B) regulate its growth.
- (C) releasing phytohormones.
- (D) protect from grazers.

Ans- (D)

24. Name the part 'X' in the given diagram. []



- (A) Acrosome

- (B) Head
- (C) Nucleus
- (D) Tail

Ans – (B)

25. In which an organism, budding occurs? []

- (A) Yeast
- (B) Paramecium
- (C) Earthworm
- (D) Amoeba

Ans – (A)

26. When your stomach is full and there is no need to food anymore, a hormone is secreted that suppresses hunger. Name the hormone. []

- (A) Ghrelin
- (B) Vasopressin
- (C) Leptin
- (D) Insulin

Ans – (C)

27. Find out the missing thing from the table. []

Type of pyramid	Base
Pyramid of number	Number of organism

.....?.....	Flow of energy
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- (A) Ecological pyramid.
- (B) Pyramid of energy.
- (C) Pyramid of bio-mass.
- (D) Giza pyramid.

Ans – (A)

28. Match the following.

[]

- | | |
|-----------|------------------------|
| 1) DNA | a. Father of genetics. |
| 2) Mendel | b. Natural selection. |
| 3) Darwin | c. Double helix. |

- (A) 1-a, 2-b, 3-c
- (B) 1-b, 2-c, 3-a
- (C) 1-c, 2-b, 3-a
- (D) 1-c, 2-a, 3-b

Ans – (D)

29. Which of the following statements is NOT true about the Mendel's selection of garden pea for his experiment? []

- (A) Well defined characters.
- (B) Bisexual flowers.
- (C) Predominantly self fertilization.
- (D) Cheap in cost.

Ans – (C)

30. The cruel campaign against which bird is initiated in 1958 in China? []

- (A) Parrots
- (B) Sparrows
- (C) Crows
- (D) Vultures

Ans – (B)

31. This symbol denotes that []



- (A) UNDP
- (B) Percolation pits
- (C) Sustainable development
- (D) Recycling logo

Ans – (C)

32. Ban all pesticides, this means that []

- (A) Promote eco-friendly agricultural practices.
- (B) Prevention of pesticides.
- (C) Control on usage of pesticides.
- (D) Stop bio-chemical factories.

Ans – (A)

33. Which of the following statement is true?

[]

(a) Development is necessary.

(b) Development should be eco-friendly.

(A) 'a' true, 'b' false

(B) 'a' false, 'b' true

(C) both a, b are false.

(D) both a, b are true.

Ans – (D)